

Multiple Choice (10 marks)

1. What is the units digit in the standard decimal expansion of the number 7^{25} ?
 - (a) 1
 - (b) 3
 - (c) 5
 - (d) 7
2. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. $x^3 + 2x + 2$ in $\mathbb{Z}_7$
 - (a) 1
 - (b) 2
 - (c) 2,3
 - (d) 6
3. If $f(z)$ is an analytic function that maps the entire finite complex plane into the real axis, then the imaginary axis must be mapped onto
 - (a) the entire real axis
 - (b) a point
 - (c) a ray
 - (d) an open finite interval
4. In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10, the identity element is
 - (a) 6
 - (b) 8
 - (c) 4
 - (d) 2
5. Let $(\mathbb{Z}_{10}, +, \cdot)$ be the ring of integers modulo 10, and let S be the subset of $\mathbb{Z}_{10}$ represented by $\{0, 2, 4, 6, 8\}$. Which of the following statements is FALSE?
 - (a) $(S, +, \cdot)$ is closed under addition modulo 10.
 - (b) $(S, +, \cdot)$ is closed under multiplication modulo 10.
 - (c) $(S, +, \cdot)$ has an identity under addition modulo 10.
 - (d) $(S, +, \cdot)$ has no identity under multiplication modulo 10.

True / False (10 marks)

Answer True or False

6. True or False: Every nonzero free abelian group has a finite number of bases.
7. True or False: The symmetric group S_3 is cyclic, meaning it can be generated by a single element.
8. True or False: The homomorphic image of a cyclic group is always cyclic.
9. True or False: The probability that a randomly chosen real number x from $[0, 3]$ is less than a randomly chosen real number y from $[0, 4]$ is $5/8$.
10. True or False: There are 6 structurally distinct Abelian groups of order 72.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Expand the following expression: $3(8x^2 - 2x + 1)$.
12. Discuss the implications of linear independence among column vectors of a 5×5 real matrix M . Identify one condition that does not relate to the others and explain why.

End of Paper